

Comparative Analysis of Depreciation Methods

Straight-Line Method vs Diminishing Balance Method

Depreciation Calculator Project

Using Excel-based Computation and Analysis

Project Background

- Businesses invest heavily in fixed assets such as machinery, vehicles, and equipment
- These assets lose value over time due to usage and technological changes
- Depreciation helps allocate asset cost systematically over its useful life

What is Depreciation?

Depreciation is the gradual reduction in the value of a tangible fixed asset

- It represents wear and tear, usage, passage of time, and obsolescence
- It is treated as a **non-cash expense** in accounting

Importance of Depreciation

- Ensures accurate calculation of business profit
- Helps match asset cost with revenue generated
- Provides tax benefits as depreciation is deductible
- Reflects true asset value in the balance sheet

Real-Time Use of Depreciation

- Manufacturing firms depreciate machinery annually
- IT companies depreciate laptops, servers, and hardware
- Transport companies depreciate vehicles
- Offices depreciate furniture and equipment

Objectives of the Project

- To understand depreciation concepts
- To study straight-line and diminishing balance methods
- To perform depreciation calculations using Excel
- To compare financial impact of both methods

Key Terminologies

- **Asset Cost:** Purchase price of the asset
- **Additional Cost:** Installation, transportation, setup costs
- **Asset Price:** Total cost of the asset (Asset Cost + Additional Cost)

Key Terminologies

- **Scrap Value:** Estimated value at the end of useful life
- **Life Span:** Number of years the asset is expected to be used
- **Book Value:** Asset value after depreciation

Straight-Line Method

- Depreciation is charged equally every year
- Same depreciation amount throughout the asset's life
- Simple to calculate and widely used

Straight-Line Method Formula

Annual Depreciation =
$$(\text{Asset Price} - \text{Scrap Value}) / \text{Life Span}$$

- Produces constant yearly depreciation
- Asset value reduces uniformly

Straight-Line Method Example

- Asset Price = ₹5,00,000
- Scrap Value = ₹50,000
- Life Span = 10 years

Annual Depreciation = ₹45,000

Straight-Line Results & Interpretation

- Total Depreciation = ₹4,50,000
- Book Value after life span = ₹50,000
- Asset is fully depreciated up to scrap value

Diminishing Balance Method

- Depreciation charged on reducing book value
- Higher depreciation in early years
- Lower depreciation in later years

Diminishing Balance Formula

Depreciation =

Opening Book Value $\times$ Depreciation Rate

- Book value changes every year
- Depreciation amount reduces annually

Rate of Depreciation

- Rate derived from straight-line method
- Used consistently every year
- Rate used in project = **20.57%**

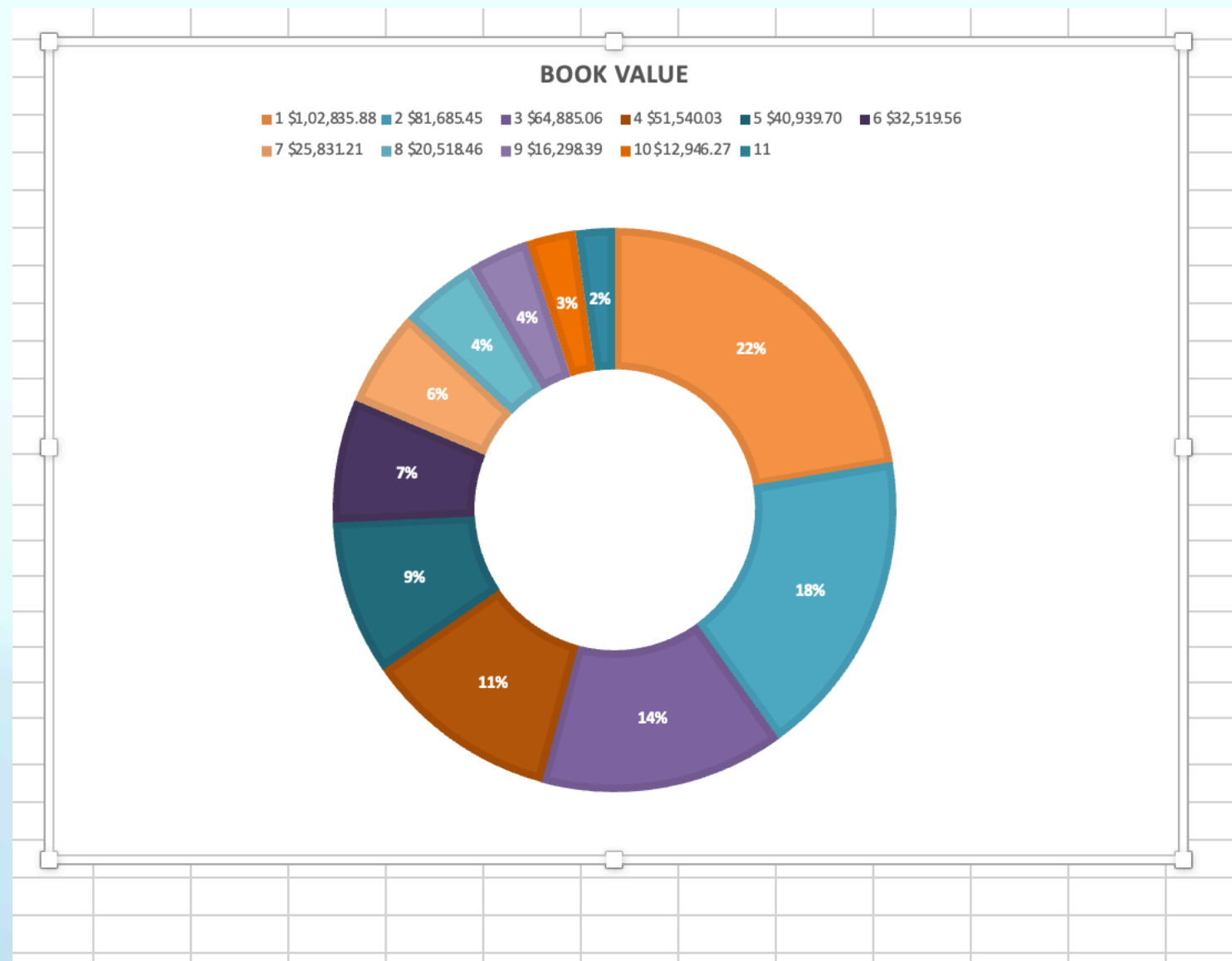
Diminishing Balance Example

- Asset Price = ₹5,00,000
- Rate = 20.57%
- Year 1 depreciation is highest
- Depreciation reduces year by year

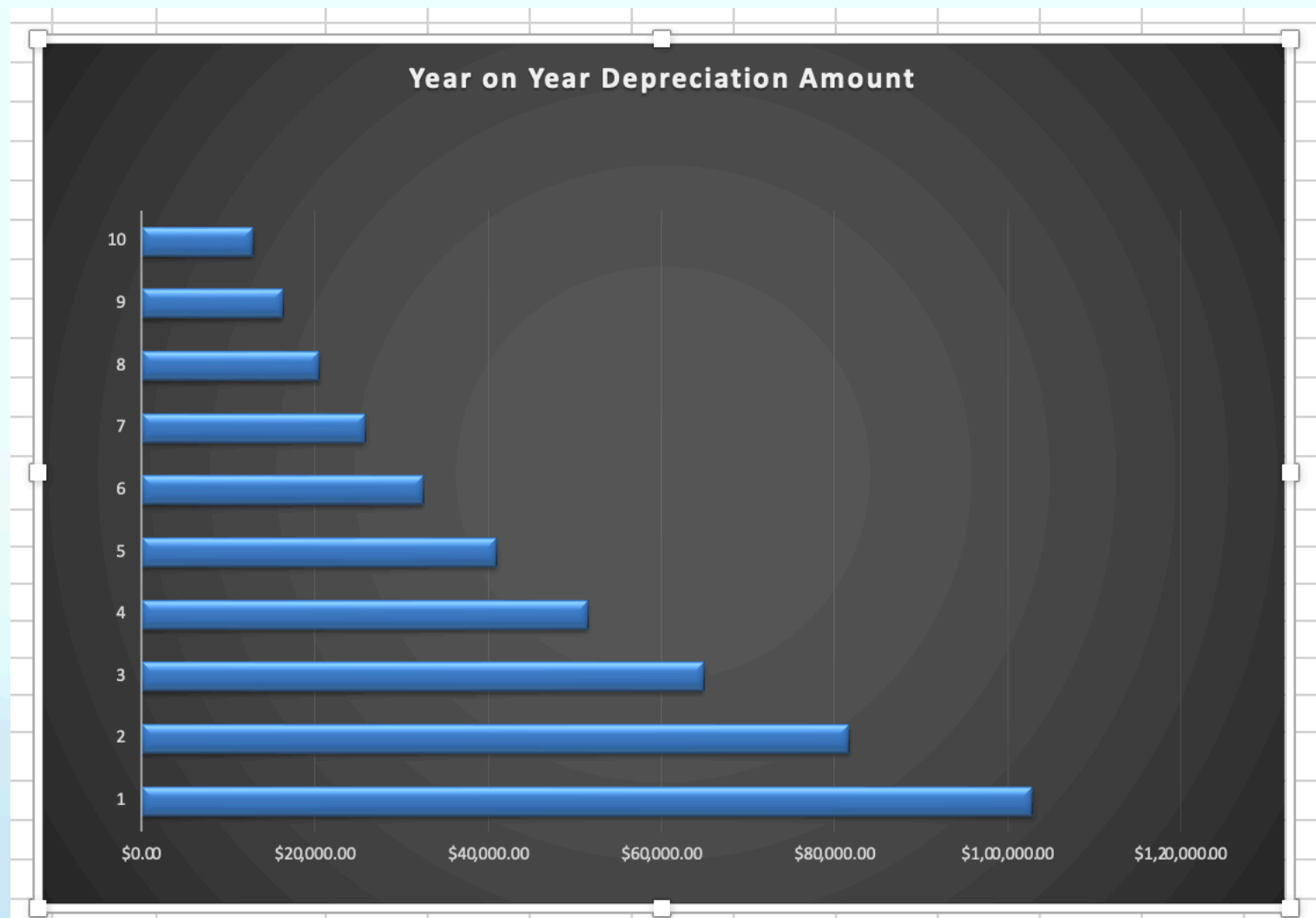
Year-wise Depreciation Schedule

- Depreciation calculated for 10 years
- Each year depends on previous book value
- Final book value after 10 years = ₹**62,946.27**

Graph: Diminishing Balance Method



Year-on-Year Depreciation Graph



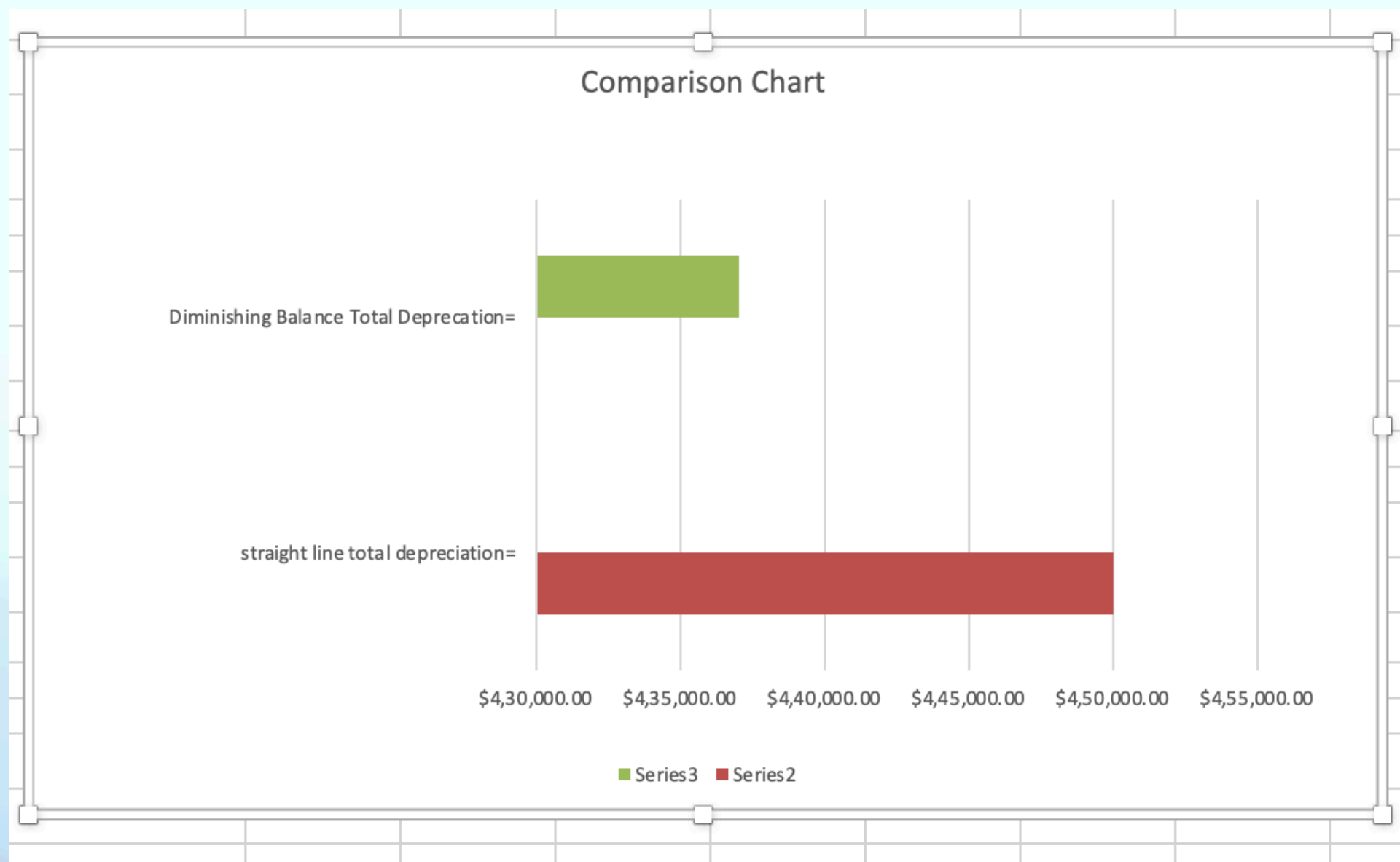
Total Depreciation (Diminishing Balance)

- **Total Depreciation =**
Asset Price – Final Book Value
 $= ₹5,00,000 - ₹62,946.27$
 $= ₹4,37,053.73$

Comparison of Total Depreciation

Method	Total Depreciation
Straight-Line	\$4,50,000.00
Diminishing Balance	\$4,37,053.73

Comparison Chart



Real-Time Business Application

- **Straight-Line:**

Buildings, furniture, long-term assets

- **Diminishing Balance:**

Computers, machinery, vehicles

Conclusion

- Straight-line method results in higher total depreciation
- Diminishing balance method better reflects actual usage
- Choice of method depends on asset nature and business strategy